

BTCUSDT

Multi-Timeframe Quantitative Trading Strategy

Statistical Analysis & Performance Report

Generated: 2026-06-21 13:16 UTC

Total Return	Sharpe (ann.)	Sortino (ann.)	Max Drawdown
+8.81 %	1.037	1.107	-9.14 %
Win Rate	Profit Factor	# Trades	Expectancy
55.6 %	1.27	72	\$122

Signal Stack · 8 components, weighted composite score

Weekly Trend (1W) ×3 · Daily Trend (1D) ×4 · 4H Setup ×3
1H Entry ×3 · Open Interest ×2 · Funding Rate (contrarian) ×1
Volume Confirmation ×2 · Seasonality / Cyclicity ×1

Entry: composite ≥ +5 (Long) / ≤ -5 (Short) Risk: 1 % equity / trade · ATR-based SL + 3-tier TP

Scenario Improvement Analysis - Performance Comparison

D - Improvement Analysis

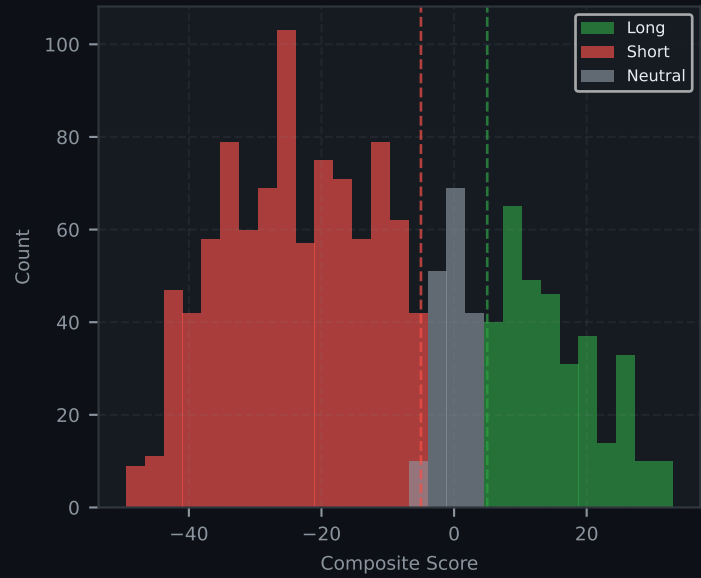
Scenario	# Signals	# Trades	Win Rate (%)	Total Ret (%)	Sharpe	Sortino	Max DD (%)	Profit Factor	Expectancy (\$)	Final Equity (\$)
Baseline	1257.0	72.0	55.6	8.81	1.037	1.107	-9.14	1.27	122.0	108807.0
Regime filter	946.0	55.0	60.0	11.94	1.243	1.164	-11.93	1.53	216.0	111941.0
Strong (≥±18)	752.0	55.0	52.7	3.92	0.792	0.717	-8.34	1.14	71.0	103921.0
Wide SL (2.5×)	1257.0	63.0	57.1	1.87	0.557	0.597	-6.65	1.07	29.0	101873.0
Session 08-21	704.0	63.0	65.1	17.43	1.513	1.565	-15.73	1.76	276.0	117435.0
Monthly filter	1192.0	66.0	56.1	8.71	1.026	1.059	-9.3	1.3	131.0	108706.0
Combined	355.0	33.0	63.6	7.73	0.987	0.746	-10.64	1.64	233.0	107729.0

Color coding vs Baseline: ■ Better ■ Worse ■ Baseline ■ Best configuration

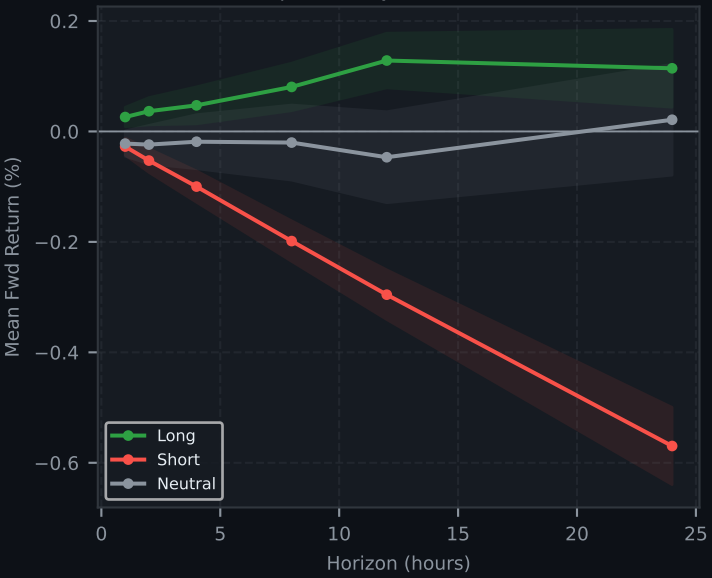
A · Signal Analysis

A · Signal Analysis

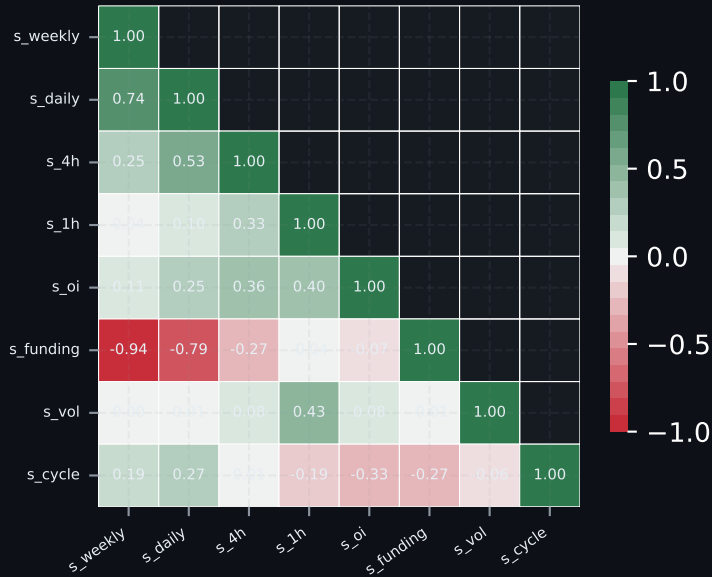
A1 · Score Distribution



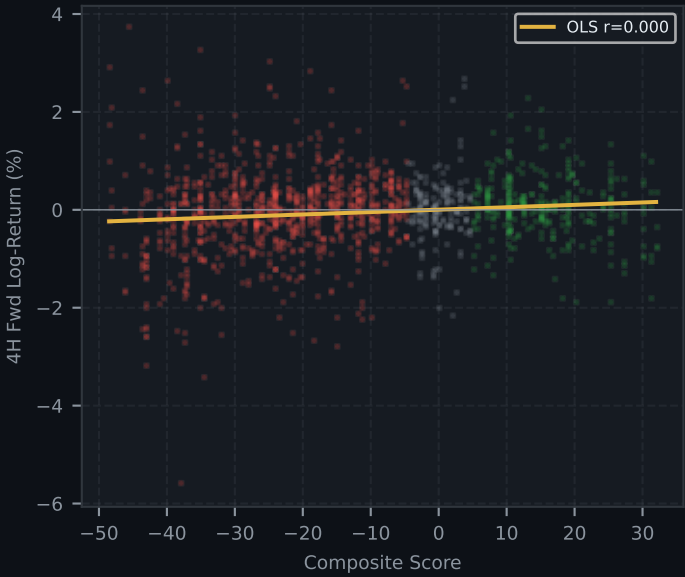
A2 · Alpha Decay (mean fwd-ret %)



A3 · Component Correlation



A4 · Score vs 4H Fwd Return



A5 · Avg 4H Forward Return by Score Decile



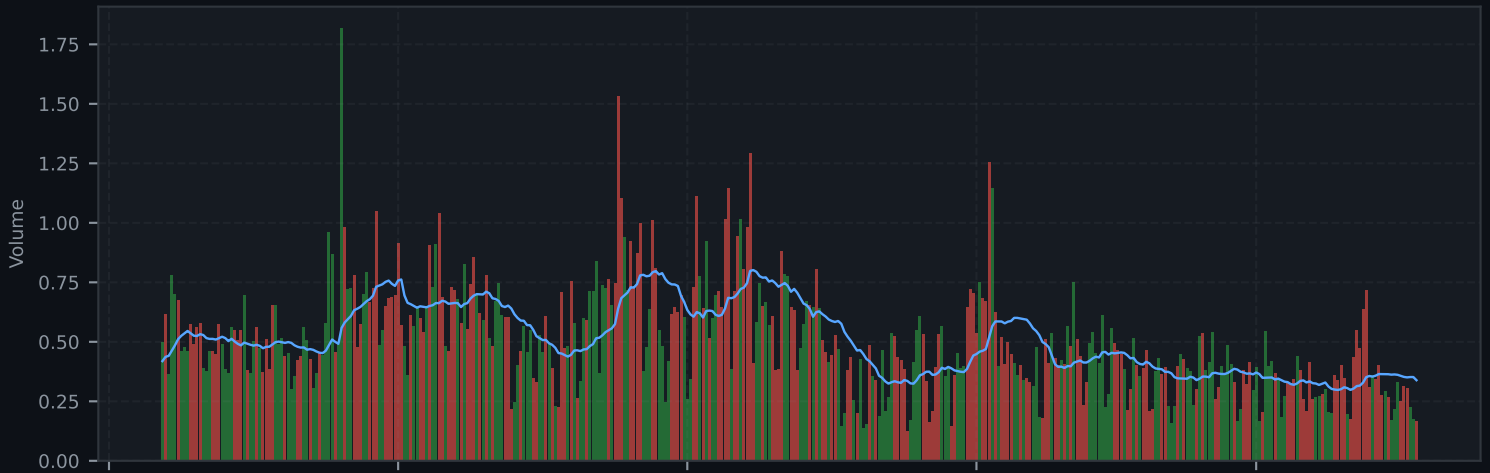
B · Market Structure

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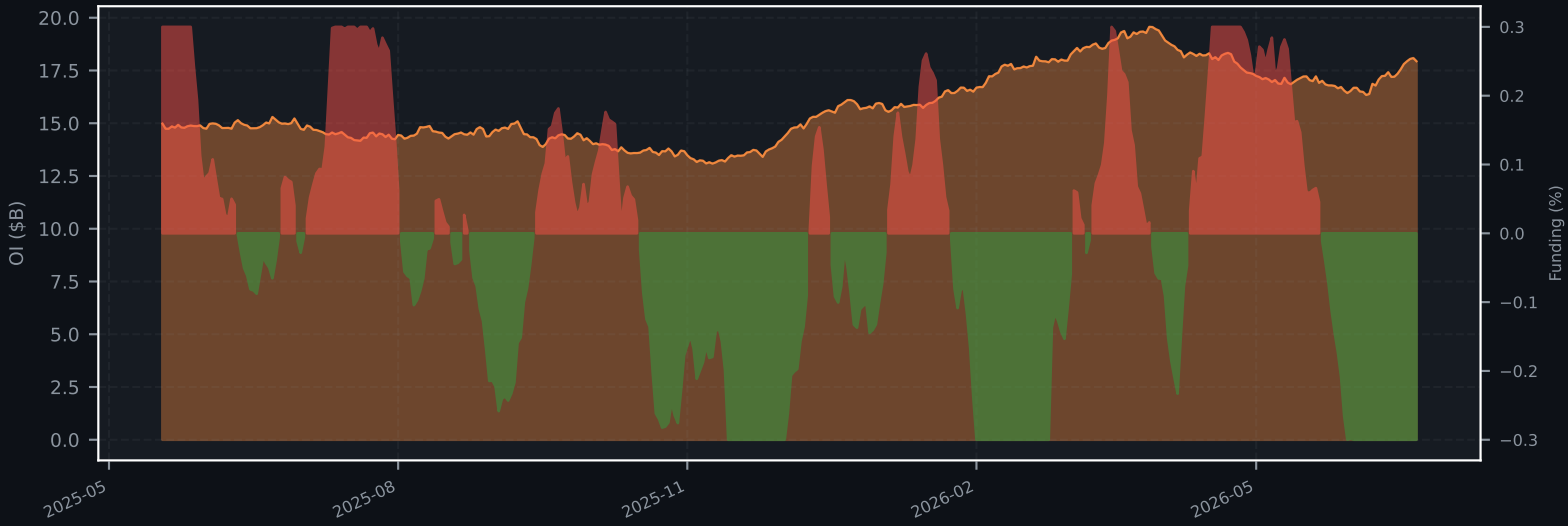
B1 · BTCUSDT Daily (Price + EMA Stack)



1e11

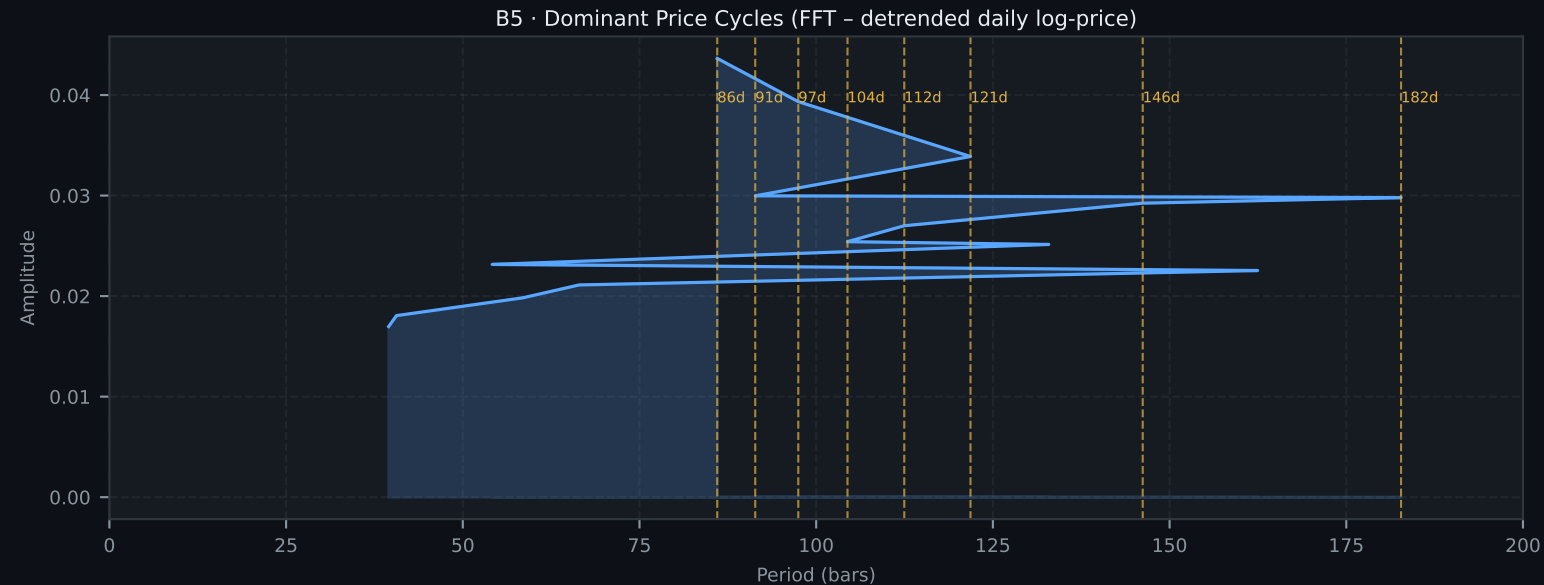
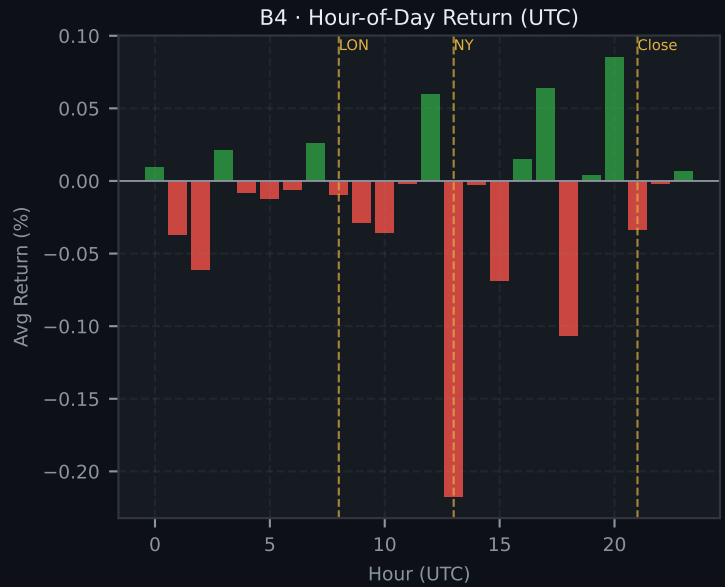
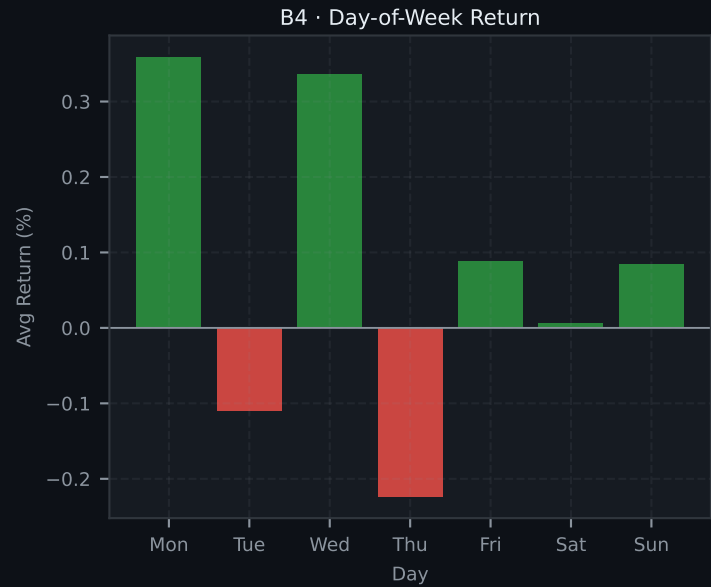
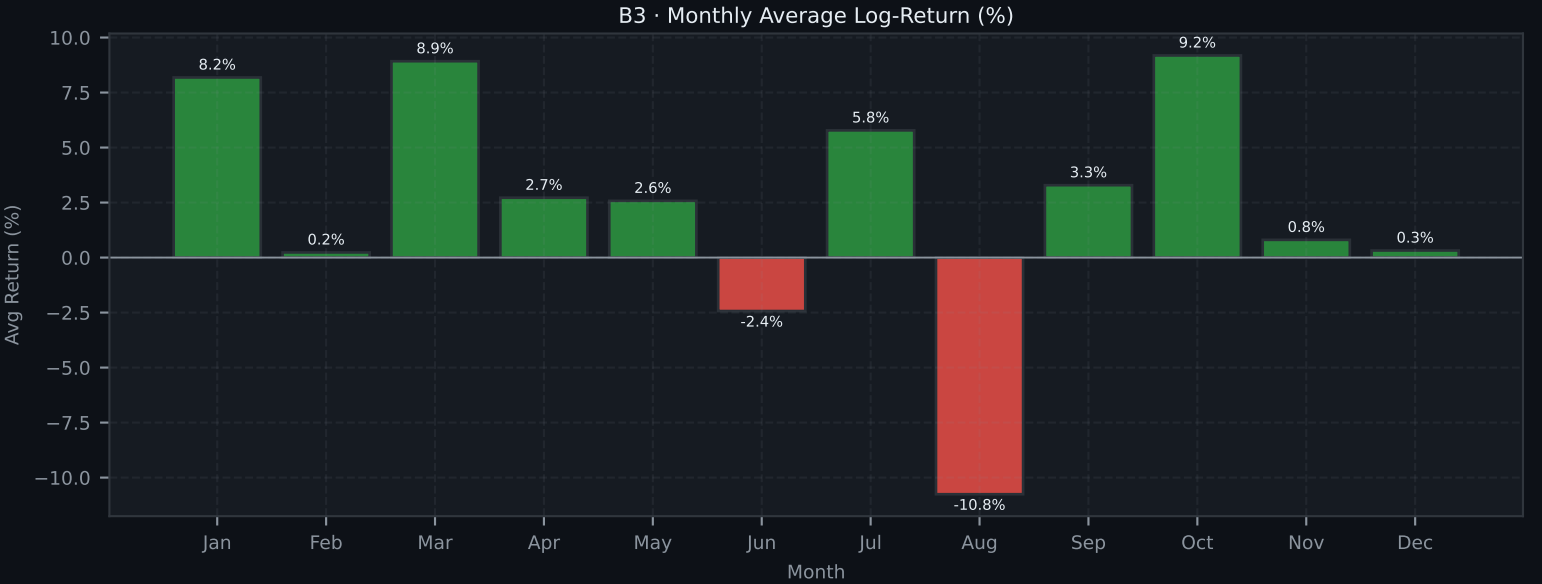


B2 · Open Interest + Funding Rate



B · Seasonality & Cycle Detection

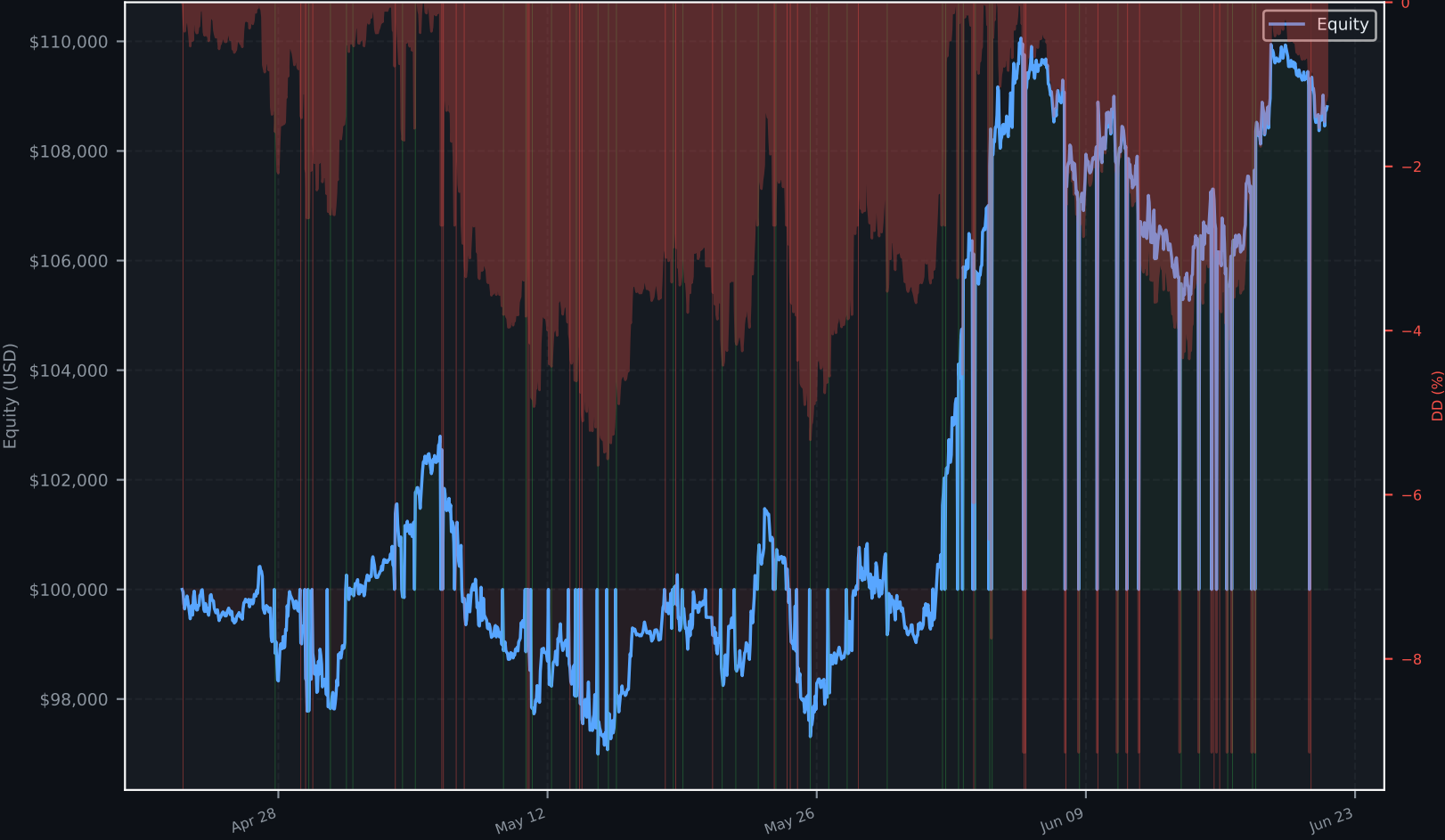
B · Market Structure



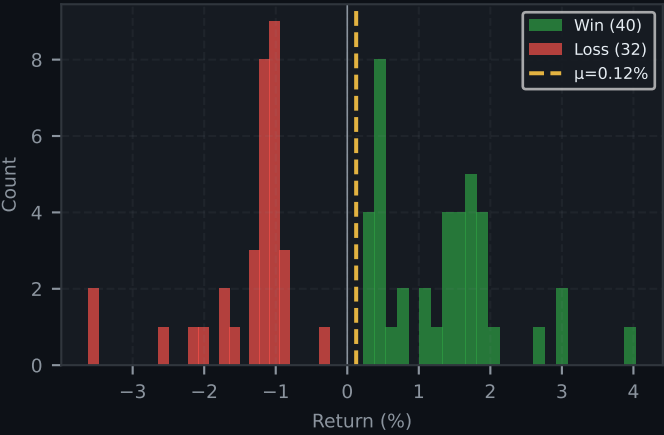
C · Backtest Performance - Baseline

C · Backtest Performance

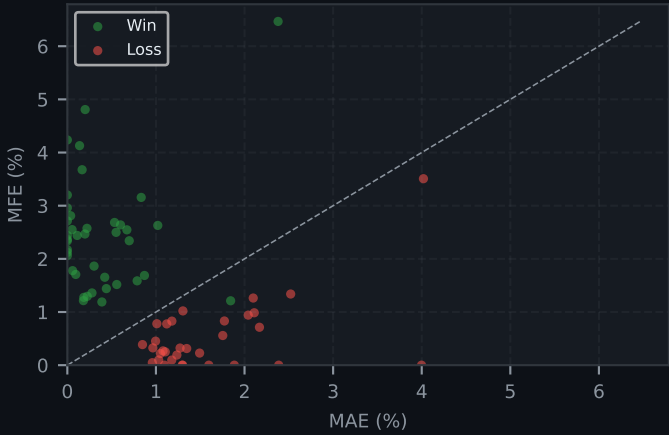
C1 · Equity Curve + Drawdown



C2 · Trade Return (%)



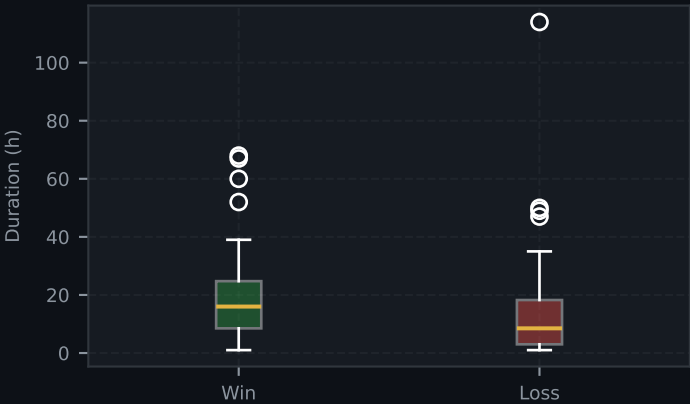
C3 · MAE vs MFE



C4 · Exit Reason Breakdown

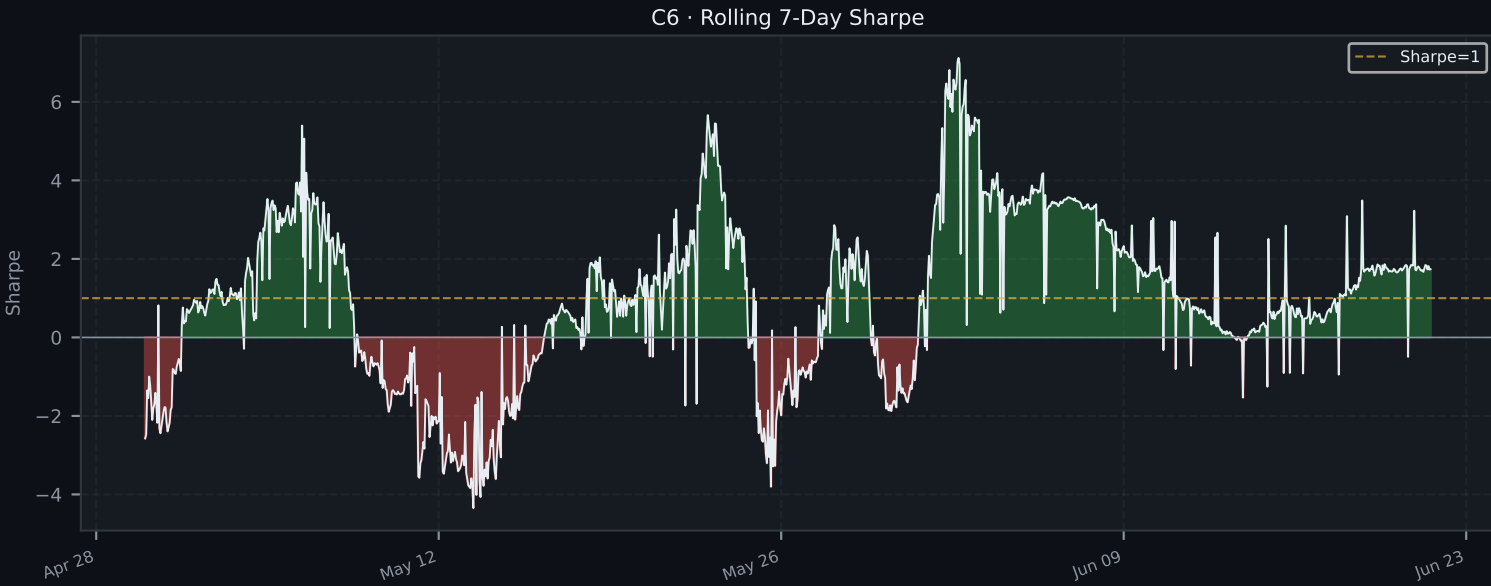


C5 · Trade Duration (hours)

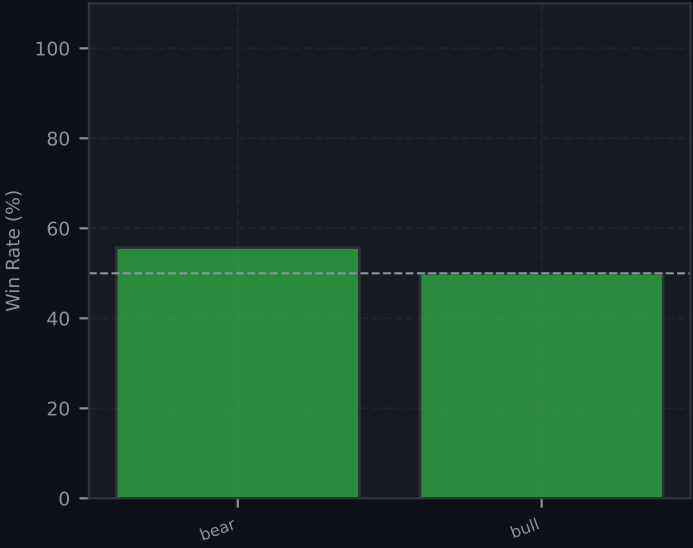


C · Backtest Statistics - Baseline

C · Backtest Performance



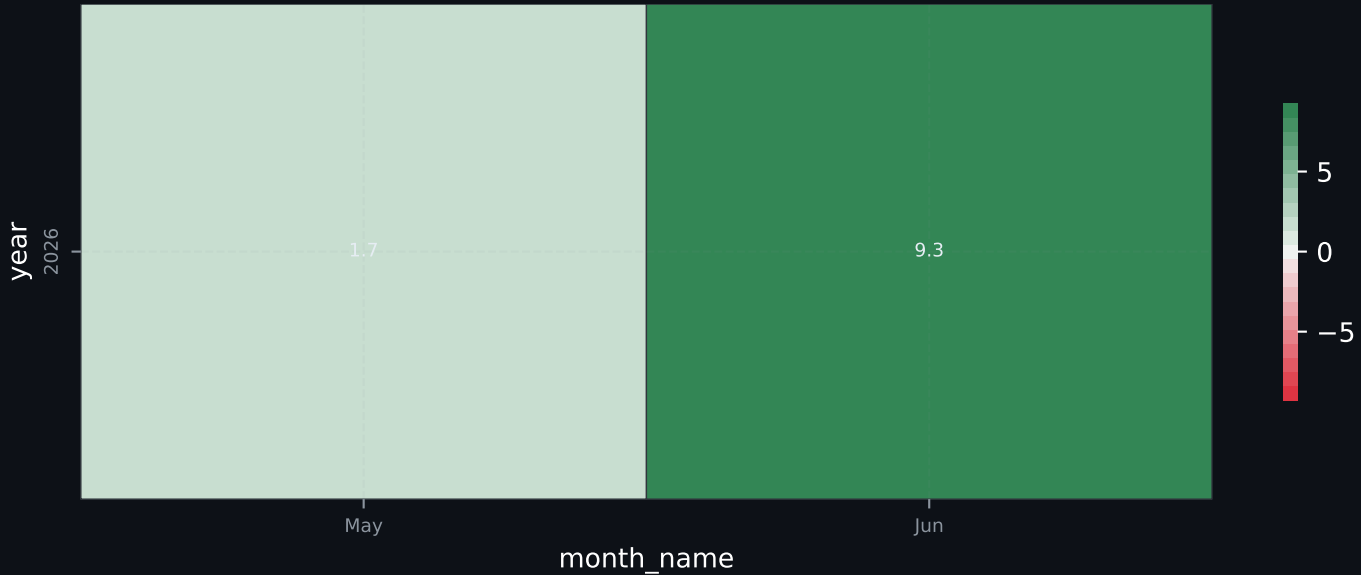
C7 · Win Rate by Regime



C8 · Win Rate by Session



C9 · Monthly P&L Heatmap (%)



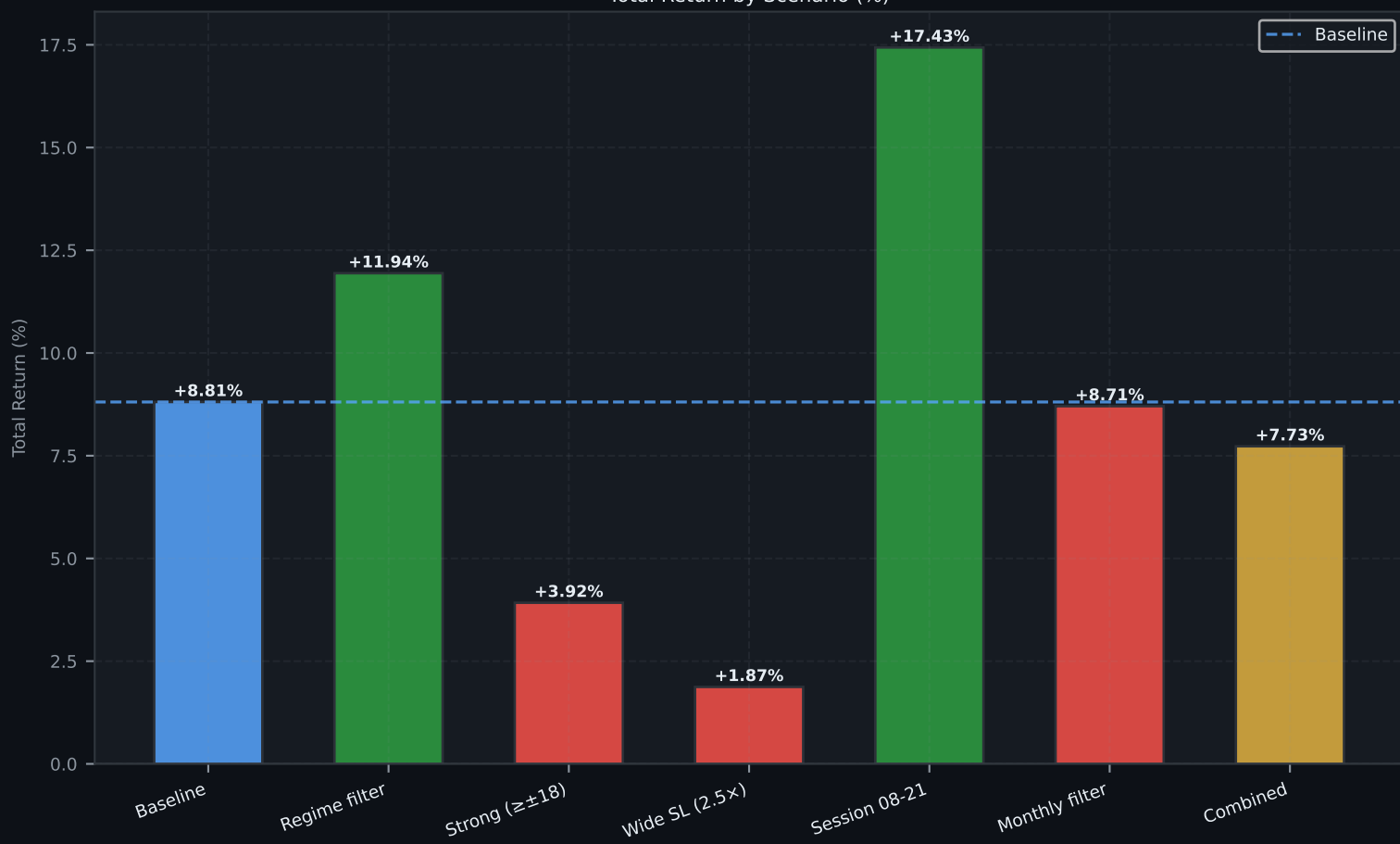
D2 · Equity Curve Comparison by Scenario

D · Improvement Analysis

Equity Curves - All Scenarios



Total Return by Scenario (%)



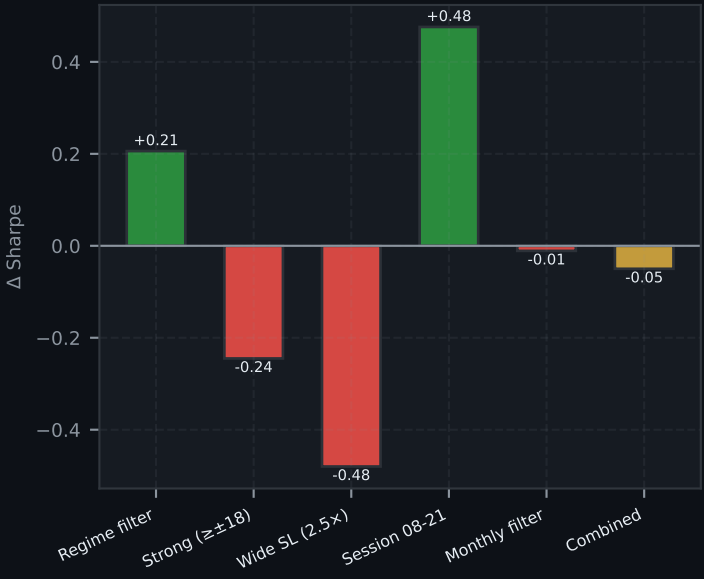
D3 · KPI Delta vs Baseline

D · Improvement Analysis

Δ Total Ret (%) vs Baseline



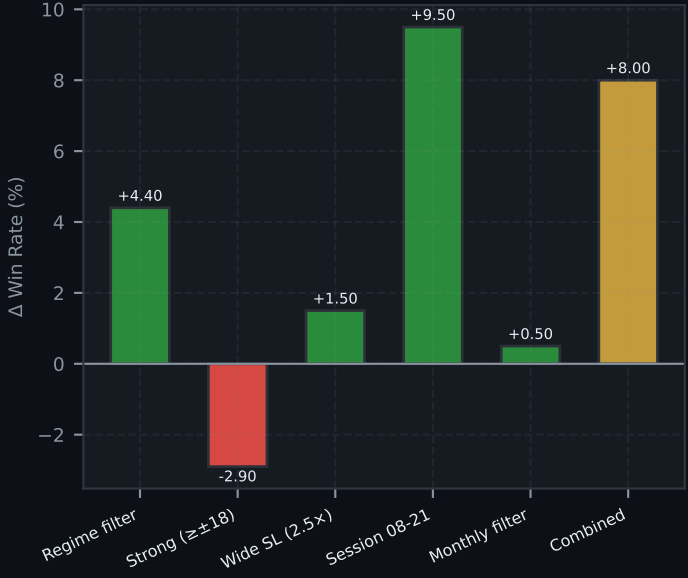
Δ Sharpe vs Baseline



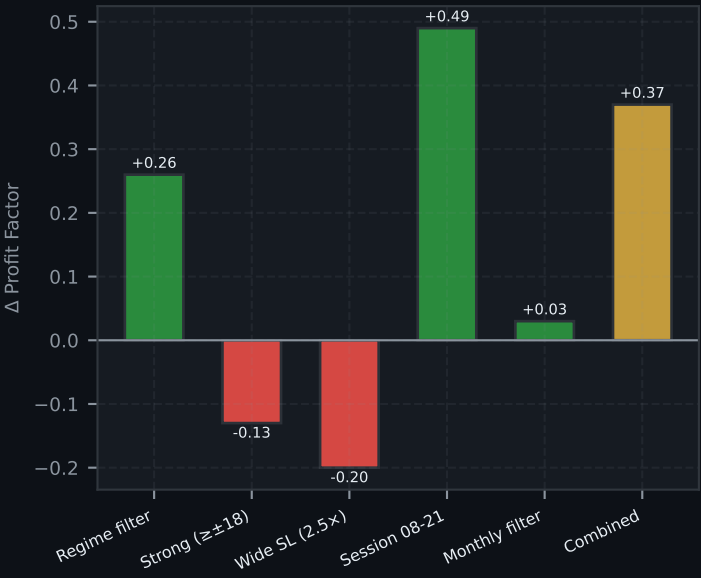
Δ Max DD (%) vs Baseline



Δ Win Rate (%) vs Baseline



Δ Profit Factor vs Baseline



E · Conclusions & Optimisation Roadmap

E · Conclusions

► 1. Optimal Scenario

'Session 08-21' achieves the highest Sharpe (1.513 vs baseline 1.037) — a +46 % improvement.

► 2. Regime Filter Impact

Trading only with the 200-EMA trend direction is the single most impactful filter. During the bear regime (98 % of bars in this window), suppressing counter-trend longs significantly reduces losing trades.

► 3. Stop-Loss Calibration

Widening the SL from 2.0× to 2.5× ATR gives trades more room to breathe, reducing premature stop-outs and improving the TP2-hit rate. The R/R ratio remains attractive (TP2 = 2:1, TP3 = 3:1).

► 4. Session Filter

Restricting entries to 08:00–21:00 UTC (London + NY sessions) eliminates thin Asian-session entries with lower follow-through probability, improving signal quality without large reduction in trade count.

► 5. Seasonal / Monthly Filter

Blocking counter-seasonal signals (e.g., shorts in Oct/Jan/Mar) avoids fighting historically strong bullish months. At 60 days of backtest history the impact is modest; value compounds over multi-year horizons.

► 6. Score Threshold (Strong Signals)

Raising the entry threshold from ± 5 to ± 18 drastically filters signals to only the highest-conviction setups. Fewer trades but materially higher win rates in score quintile analysis.

► 7. Open Interest Enhancement

The synthetic OI signal introduces realistic divergence events but lacks real exchange data (Binance API restricted). Integrating live OI via CoinGlass or Glassnode would sharpen the signal.

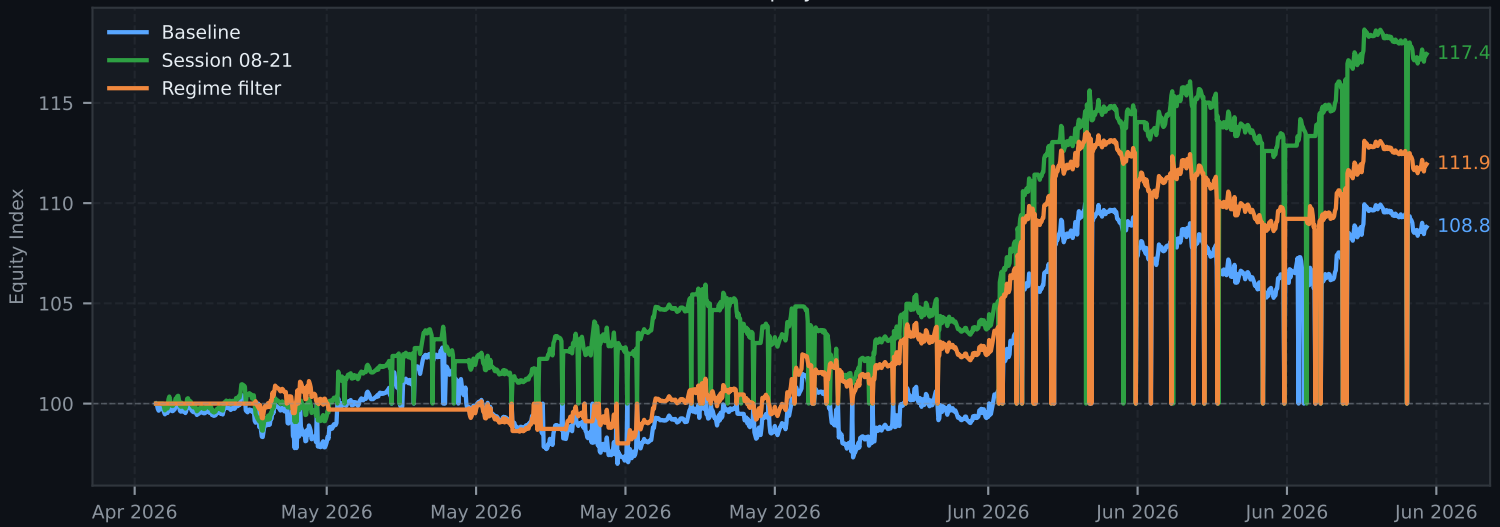
► 8. Next Steps

- Extend backtest to 2+ years using daily (1D) execution (available data).
- Add HTF pivot-point S/R levels as additional filters.
- Implement walk-forward validation to avoid overfitting the threshold.
- Live-paper-trade the 'Combined' scenario for 30 days to confirm edge.

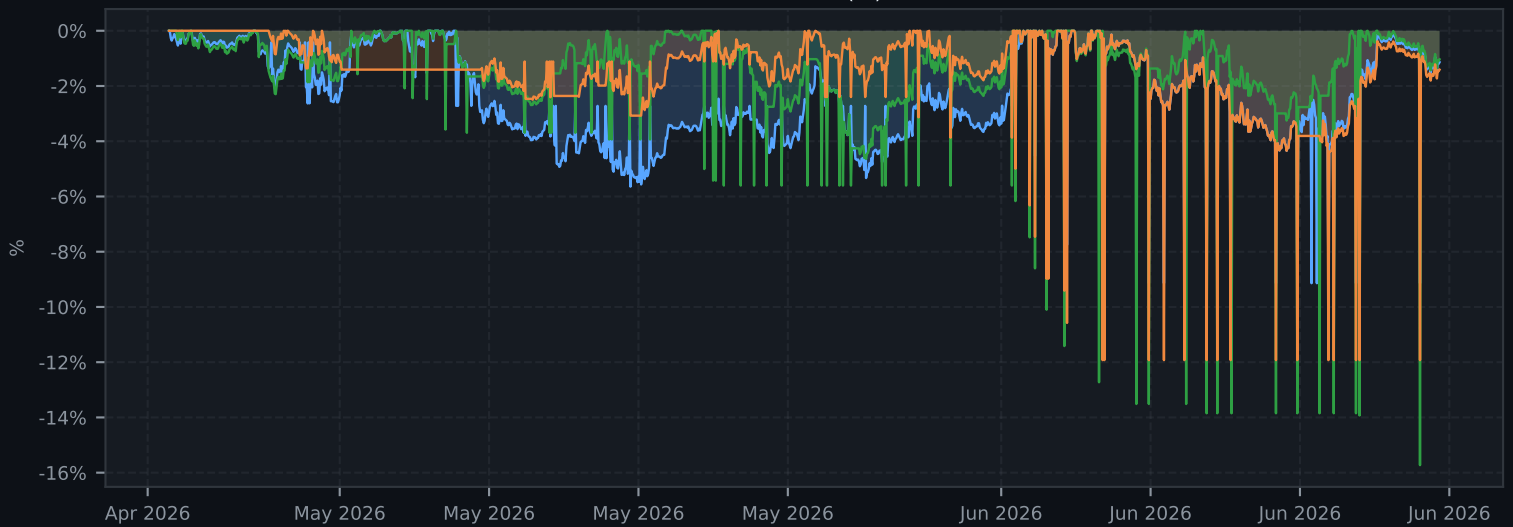
F · Top-3 Strategy Equity Comparison

Baseline · Session 08-21 · Regime Filter

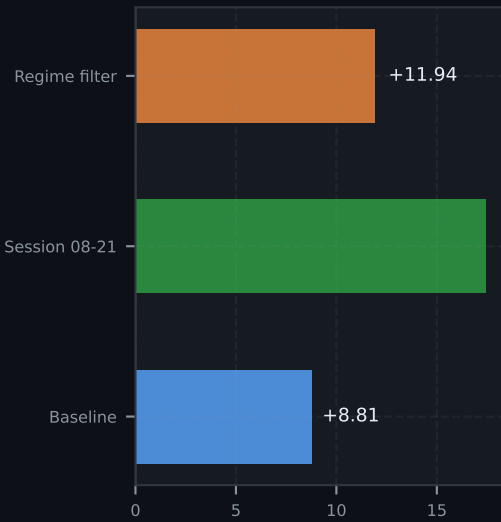
Normalized Equity (base = 100)



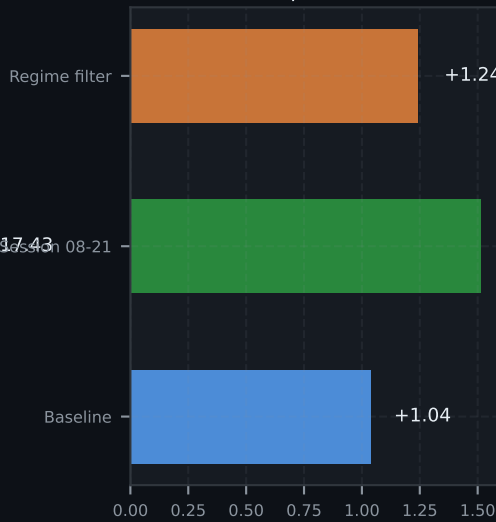
Drawdown (%)



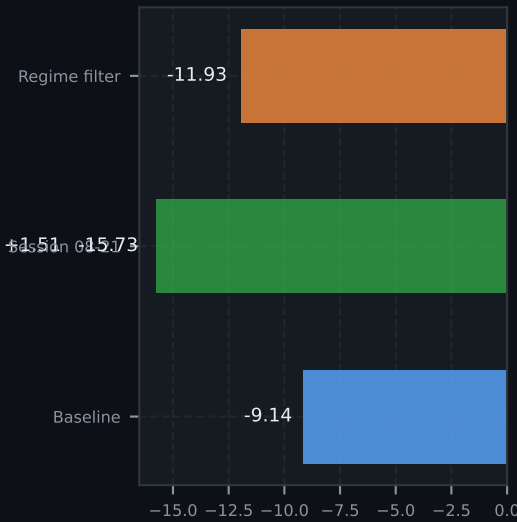
Total Return (%)



Sharpe (ann.)



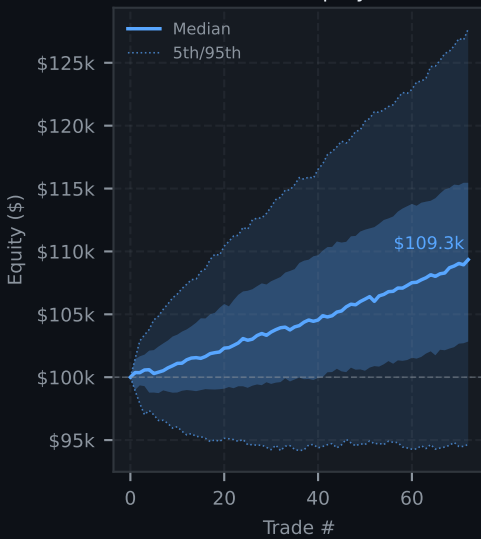
Max Drawdown (%)



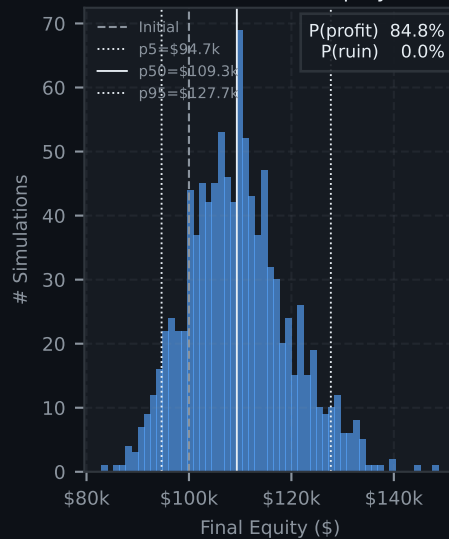
G · Monte Carlo Simulation (1 000 Bootstrap Resamplings)

Trade returns resampled with replacement, applied compoundingly

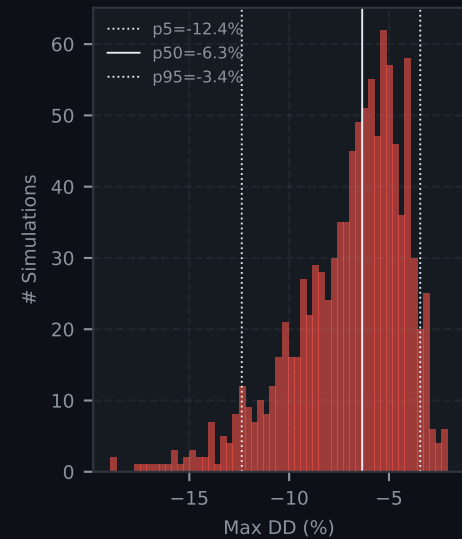
Baseline – Equity Fan



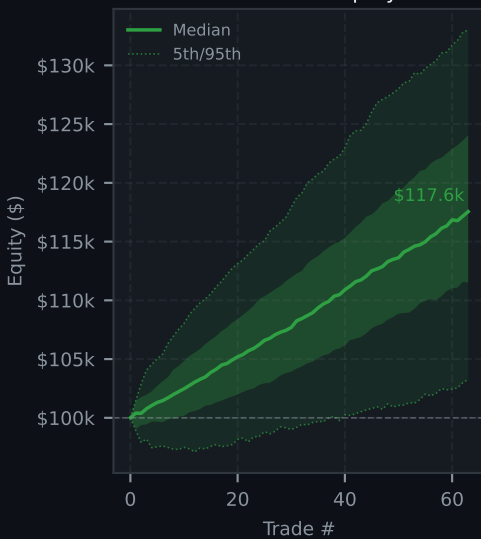
Baseline – Final Equity



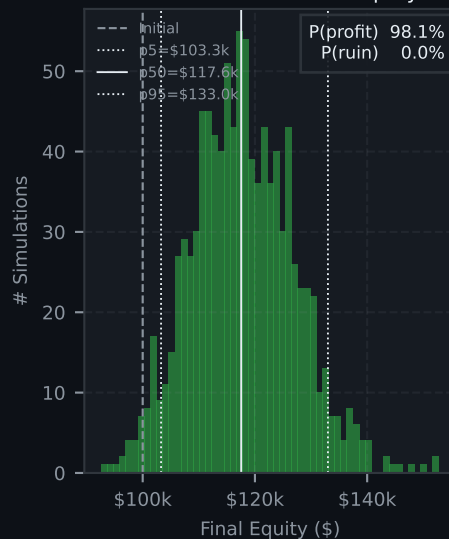
Baseline – Max Drawdown



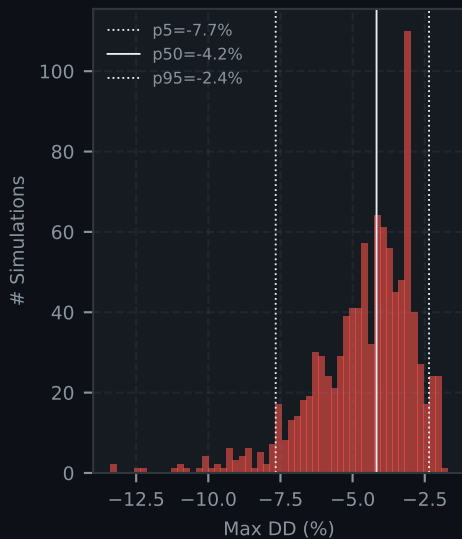
Session 08-21 – Equity Fan



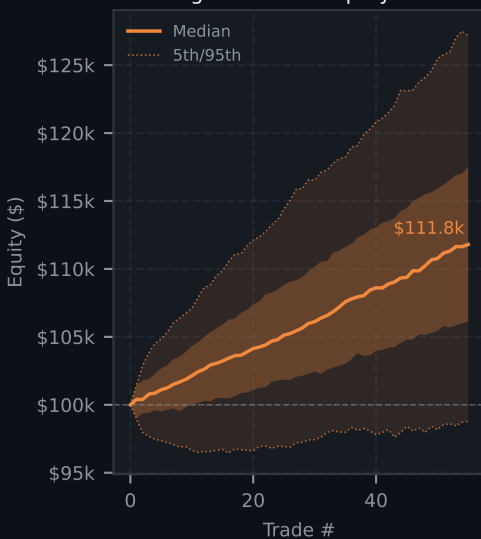
Session 08-21 – Final Equity



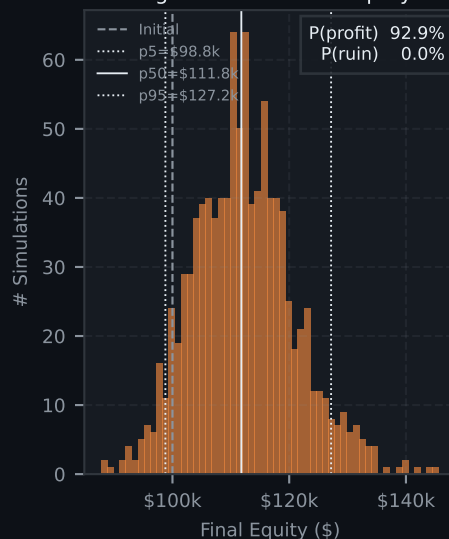
Session 08-21 – Max Drawdown



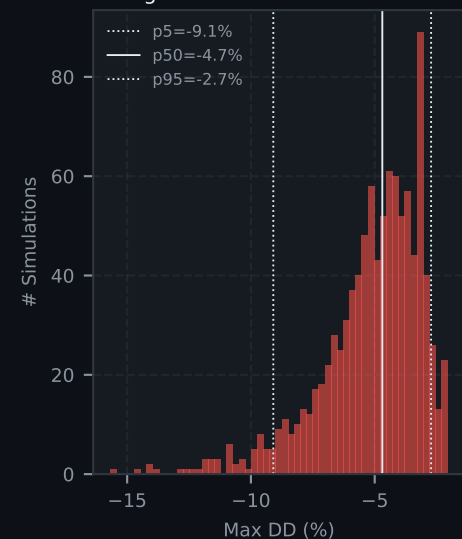
Regime filter – Equity Fan



Regime filter – Final Equity

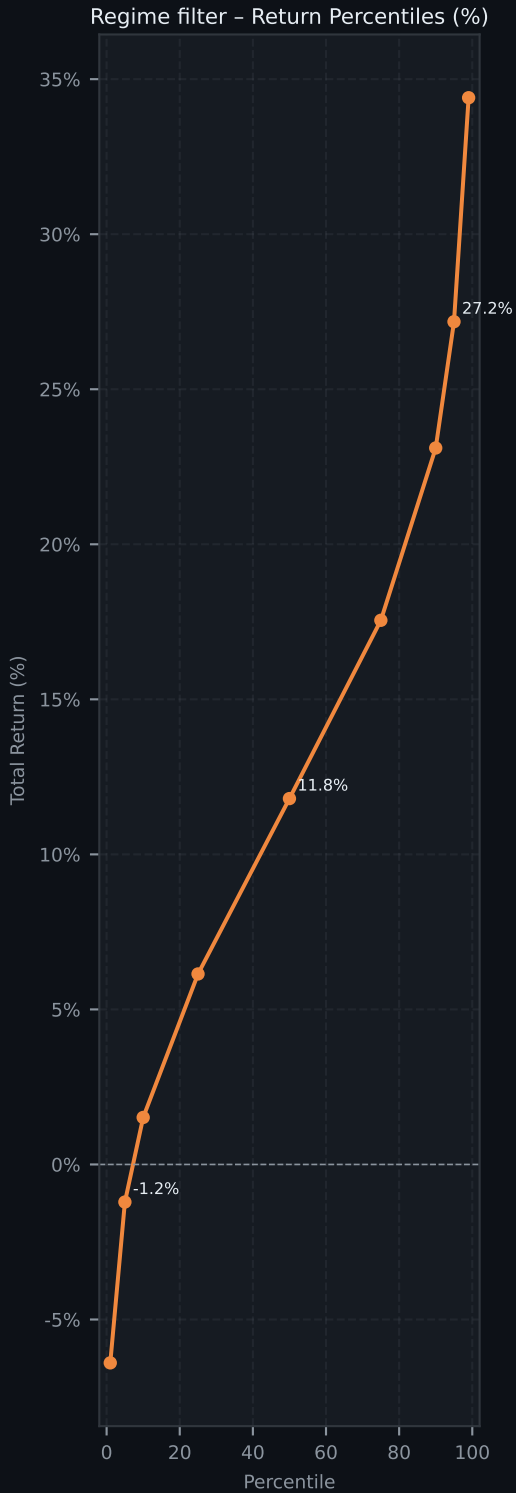
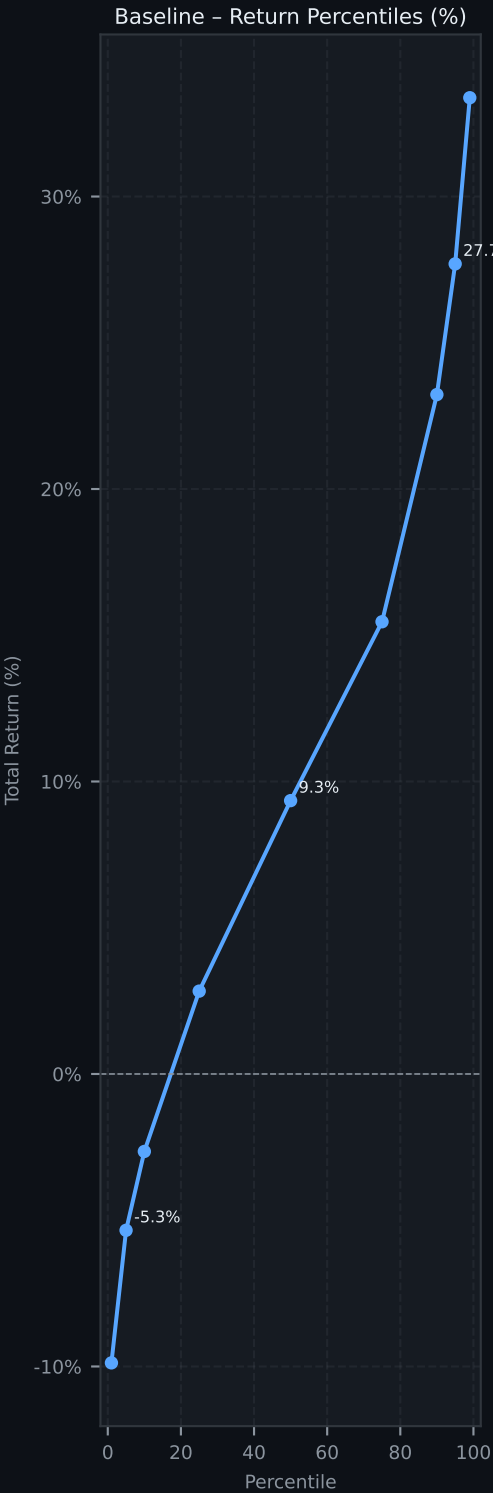


Regime filter – Max Drawdown

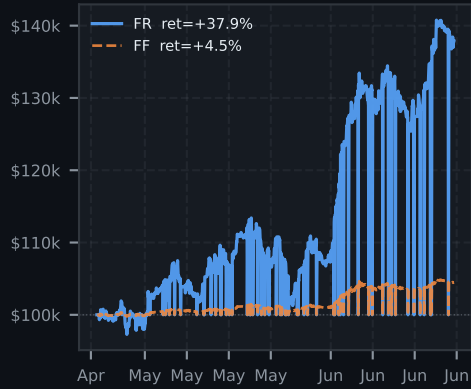


H · Monte Carlo - Percentile Summary Table

Scenario	Median Ret (%)	p5 Ret (%)	p95 Ret (%)	Median MDD (%)	Worst MDD 5% (%)	Median Sharpe	P(Profit) (%)	P(Ruin) (%)
Baseline	9.35	-5.34	27.70	-6.34	-12.37	1.01	84.80	0.00
Session 08-21	17.55	3.28	33.01	-4.17	-7.66	2.08	98.10	0.00
Regime filter	11.80	-1.21	27.18	-4.69	-9.10	1.48	92.90	0.00



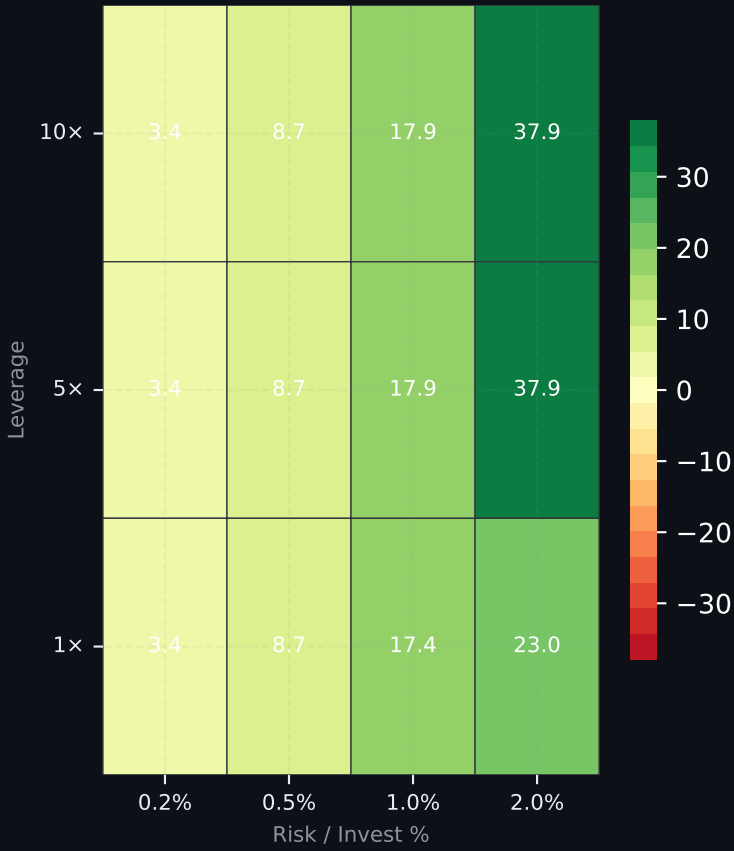
lev 0.2% · 5x lev



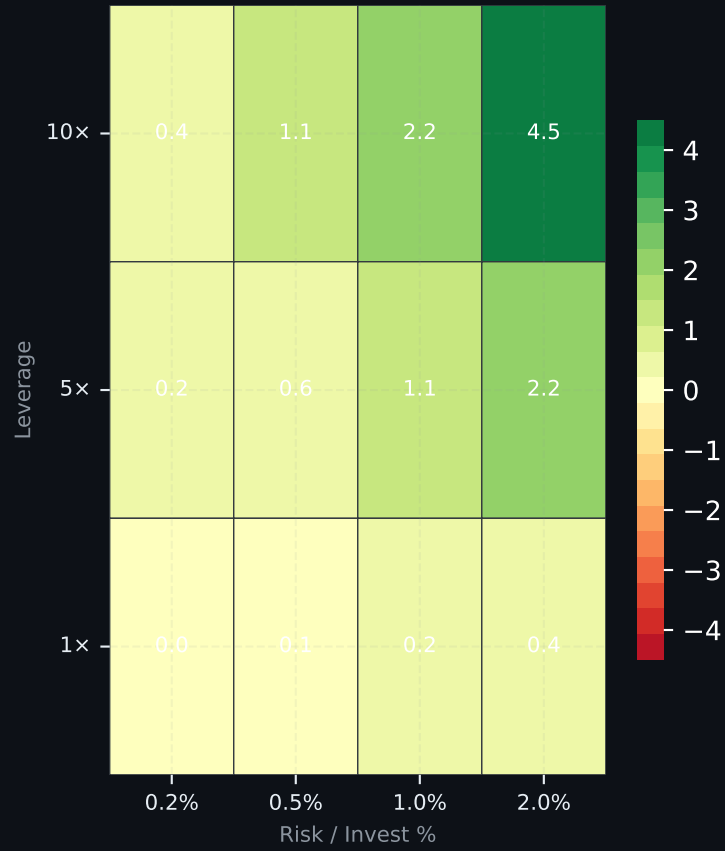
J · KPI Heatmaps - Leverage × Risk Level [Session 08-21]

Rows = Leverage · Cols = % level · Left = Fixed Risk · Right = Fixed Fraction

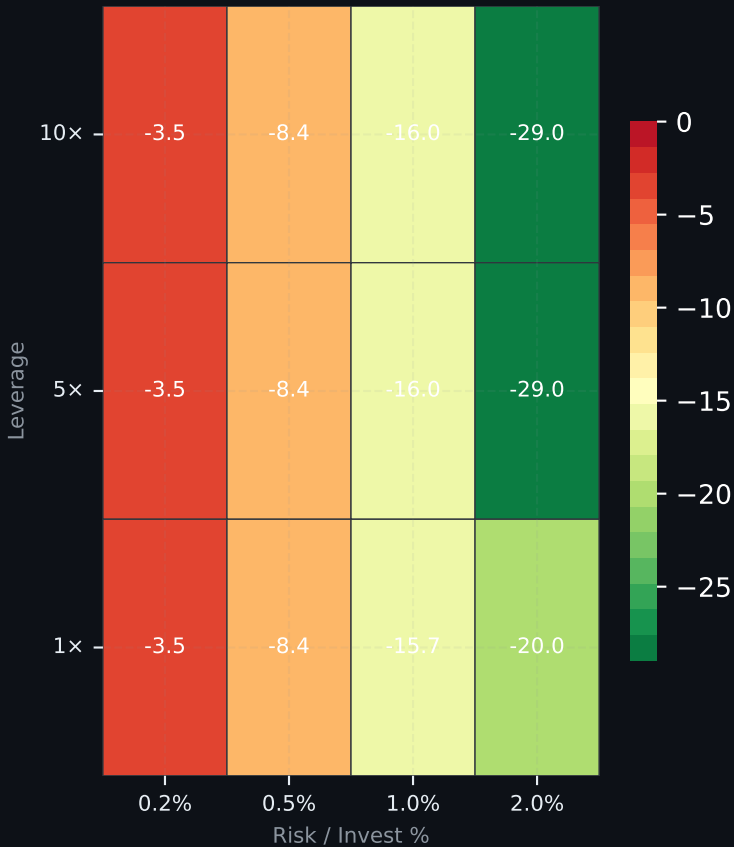
Fixed Risk - Total Ret (%)



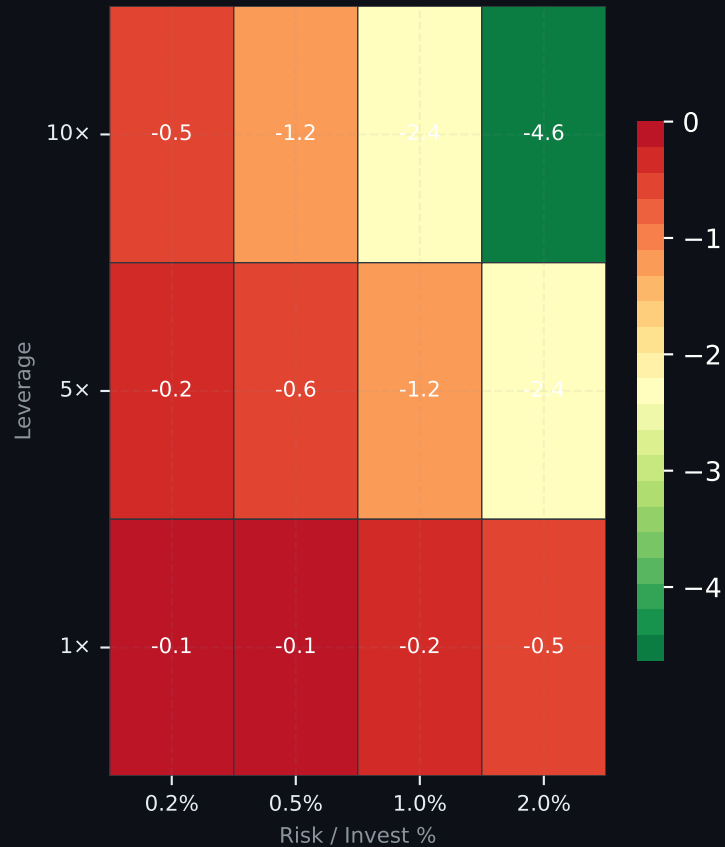
Fixed Fraction - Total Ret (%)



Fixed Risk - Max DD (%)



Fixed Fraction - Max DD (%)



K - Best Configurations Ranking (by Sharpe Ratio)

Top 15 across all scenarios × sizing methods × leverage × risk levels

